The GTE Particle Spectrum at n=10: Structure, Laws, and Novel Predictions from Large-Scale UGP-Guided Discovery

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Abstract

We present a comprehensive analysis of the Generative Triple Evolution (GTE) particle spectrum at n=10, derived from the Universal Generative Principle (UGP). A large-scale discovery run generates 1,000,035 candidate particles below 173 GeV, classified into viability tiers (Green, Blue, Purple, Orange, Red) by a scoring system that reduces in practice to viability-only ranking (GTE compliance = 1.0 for all candidates by construction; stability score is effectively constant).

Key findings: (1) all 24 known Standard Model particles rank in the Green tier by both structural landscape position and blind composite score (before force-labeling), with 50-fold SM enrichment ($p < 10^{-4}$ vs. random draw; three distinct notions of recovery defined in §5); among the 19 candidates with confidence ≥ 0.90, all are SM particles; (2) piecewise-linear hinge laws in the $(k, \log m)$ plane with universal law-family parameters $B = 4.01 \times 10^{-6}$ and $D = -3.13 \times 10^{-6}$; (3) genuine oscillatory structure with intrinsic period ∼100,000 steps (z>50, stable across five window-size tests; a prior single-cycle estimate of ∼500,000 steps is corrected here); (4) multivariate surfaces linking ladder index, branch, and c-state to mass and lifetime; (5) 9 genuinely novel candidate states (GTE-P1, P2, P3, P6, P7, P8, P10, P11, and trajectory-reinterpreted P9) representing mass-band predictions (calibration uncertainty ±1–40% depending on candidate; three align with companion paper first-principles derivations); and (6) trajectory-path multiplicity: GTE-P4 and P5 share GTE triples with the canonical muon and electron respectively but arrive via distinct evolution paths at 137 MeV and 21 MeV — these are trajectory-path variants of SM particles, not independently novel predictions.

All results are derived from a single deterministic run with frozen code and integrity-checked artifacts (SHA-256 hashes recorded).

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1 Introduction

The Standard Model (SM) of particle physics provides an extraordinarily successful description of fundamental particles and their interactions, yet the origin of its parameter set—masses, lifetimes, mixing angles, and coupling constants—remains unexplained. The Universal Generative Principle (UGP) [8, 9] and its derived Generative Triple Evolution (GTE) provide a deterministic, arithmetic foundation from which particle families can be enumerated as integer triples at a single structural parameter n=10.

The companion paper [6] presents derivations of the SM fermion mass spectrum, gauge couplings, CKM and PMNS mixing matrices, neutrino masses, the Higgs sector, and the cosmological constant within the UGP framework. That companion paper operates at the level of *individual parameter derivation*: given a canonical triple, the Universal Calibration Law (UCL) maps it to a specific mass.

This paper addresses a complementary question: *what is the full landscape of particles that the GTE arithmetic generates at n=10?* Rather than deriving individual masses, we perform a million-scale computational survey of the entire GTE spectrum, characterizing its mathematical structure, identifying where SM particles sit within it, and extracting novel predictions from the highest-confidence non-SM candidates.

The paper makes the following contributions:

(i) A comprehensive GTE spectrum of 1,000,035 candidates, classified by a multi-component viability scoring system (§3).

(ii) Universal hinge laws and law-family parameters governing the $(k, \log m)$ plane (§4).

(iii) Genuine oscillatory structure with intrinsic period ∼100,000 steps ($z > 50$, confirmed across five window sizes, §4).

(iv) Recovery of all 24 SM particles in the highest-viability tier, with a 50-fold enrichment confirmed by a permutation null test (§5).

(v) Nine genuinely novel high-confidence predictions (GTE-P1, P2, P3, P6, P7, P8, P9, P10, P11; plus 2 trajectory-path variants GTE-P4/P5 excluded from the novel register), three of which are internally consistent with the companion paper's first-principles mass derivations (§6).

(vi) A robustness battery including FDR correction, degrees-of-freedom accounting, and explicit disclosure of calibration assumptions (§7).

1.1 Status of Claims

To prevent conflation of derived, calibrated, and open results, we classify every claim in this paper:

DS-A: Derived from GTE arithmetic.
[1] Hinge law structure; oscillation period and significance; law-family B, D parameters; trajectory-path multiplicity pattern. These are deterministic outputs of the GTE algorithm with no free parameters beyond n=10.

DS-A′: Pipeline-derived (deterministic given matching protocol).
SM particle recovery in Green tier. The 24 SM matches are identified by mass-proximity matching against PDG values and force-classified to Green. This is deterministic given the matching protocol and PDG targets but is not parameter-free in the strict sense.

DS-B: Calibrated quantities.
Predicted novel particle masses (9 genuinely novel: GTE-P1, P2, P3, P6, P7, P8, P9, P10, P11; plus trajectory-path variants GTE-P4 and P5). These are GTE-generated candidates whose masses have been corrected by a PCHIP spline calibrator anchored on 9 known SM fermion masses. The calibrator is a systematic correction tool, not a first-principles derivation. The companion paper [6] provides the independent first-principles mass derivation.

DS-C: Pipeline design choices.
Green/Blue/Orange/Red tier thresholds (percentiles of empirical viability distribution); $1\,\mu$s stability threshold; 70% theory-confidence floor. These are engineering choices for experimental prioritization, not theoretical claims.

DS-D: Open or partial.
Quantum number assignments for novel predictions; lifetime predictions for non-SM candidates; confirmation of the ∼100,000-step oscillation period beyond the tested five window sizes; longer runs ($k_{\max} \geq$ 1,000,000) are needed.

[1] "DS-" (Discovery Survey) labels are GTE particle-discovery survey categories local to this paper, distinct from the series claim-strength taxonomy ($\mathrm{A}_{\mathrm{Lean}}$, $\mathrm{A}_{\mathrm{MDL}}$, A/D, B, D; [3]). The DS prefix prevents namespace collision with the canonical taxonomy.

Significance

We report the first million-scale computational survey of the particle spectrum predicted by the Universal Generative Principle (UGP) at n=10. From a single deterministic arithmetic algorithm operating on integer triples, the Generative Triple Evolution (GTE) generates 1,000,035 candidate particles spanning the full Standard Model (SM) mass range. All 24 known SM particles rank in the highest-viability tier by both structural position in the landscape and blind composite score (before any force-labeling), with 50-fold enrichment over the random baseline ($p < 10^{-4}$). The landscape exhibits universal hinge laws, statistically robust oscillatory structure (period ∼100,000 steps, $z > 50$, provisional physical interpretation), and multivariate surface structure. We identify 9 genuinely novel high-confidence candidate states (GTE-P1, P2, P3, P6, P7, P8, P10, P11, and the reinterpreted P9) plus 2 trajectory-path variants of canonical SM particles (GTE-P4, P5, which share GTE triples with the muon and electron respectively but arrive via distinct evolution paths); all masses are calibrated predictions (DS-B) and should be understood as mass-band estimates rather than sharp predictions, except where individually calibration-stable (GTE-P1: ±1.4%; GTE-P2: ±14%; others up to ±40%). These results provide strong evidence that the GTE arithmetic at n=10 is a mathematically structured lattice whose highest-viability outputs are statistically inconsistent with a random baseline (see Fig. 8 for the complete SM particle spectrum prediction).

2 The UGP Framework at n=10

We summarize the essential elements; full details are in [8, 6].

UGP axioms. The Universal Generative Principle is an integer-based dynamical system operating on triples $(a, b, c; g)$ under axioms of locality, symmetry, and compression. At structural parameter n=10, the UGP ridge is $R = 2^{10} - 16 = 1008$, from which two admissible divisor seeds emerge:

$$(1, 73, 823; 1), \qquad (1, 73, 2137; 1).$$

These initialize the first-generation leptonic and quark families. Once n is fixed, the ridge and seeds are uniquely determined up to mirror symmetry.

GTE rules. The Generative Triple Evolution defines stepwise evolution:

- **Odd steps:** transition to ridge states ($c = 2^n - 1$).
- **Even steps:** increment b by the Fibonacci lift $F_{|q_t - q_{t-1}|}$; update a and c via latched-write rules.

At n=10, the first even-step increment is locked at $F_{13} = 233$, producing the canonical ladder. The "Mersenne jump" at $G_2 \to G_3$ replaces the naïve $bq + 15$ rule with $c_3 = 65535$. The evolution is fully deterministic: given seeds and horizon limits, all possible triples are enumerable.

Calibration. To map integer triples to physical masses, the Universal Calibration Law (UCL) computes a dimensionless calibration factor C_f from the triple coordinates. A base energy E_{base} is derived from the information content $N = |b|$, yielding $m = C_f \cdot E_{\text{base}}$. The companion paper [6] derives the UCL coefficients from first principles via the Elegant Kernel and Quarter-Lock symmetry. In this paper, masses are further corrected by a PCHIP spline interpolator trained on 9 SM fermion anchor masses (see §3).

Machine-checked foundations. The UGP ridge sieve, RSUC uniqueness classification, GTE orbit structure, and Quarter-Lock theorem are machine-checked in Lean 4 [9]. The Two-Layer PSC Theorem forces the SM gauge group $SU(3) \times SU(2) \times U(1)$ and $N_{\text{gen}} = 3$ [7].

3 Discovery Protocol and Dataset

3.1 Generation pipeline

The discovery engine (`Verifier_discovery_engine_v4.py`) implements the full GTE trajectory at n=10 with the following preset parameters:

Parameter	Value
`max_even_steps`	500,000
b_{max}	1×10^9
m_{max}	173,000 MeV (top quark mass)
Workers	14 (parallel)
GTE mode	`exact` (strict compliance)
Target sectors	fermions, neutrinos, bosons

The run produces 1,000,035 candidate particles across both the "our" and "mirror" GTE branches, plus neutrino and boson sectors.

3.2 PCHIP calibration

Raw UCL masses are corrected by a PCHIP (Piecewise Cubic Hermite Interpolating Polynomial) spline fitted on exactly 9 SM fermion anchor masses:

Particle	n-value	PDG 2020 mass (MeV)
Electron	73	0.511
Up	9	2.16
Down	5	4.67
Muon	42	105.66
Strange	186	93.0
Charm	275	1,270
Tau	275	1,777
Bottom	8,191	4,180
Top	337,920	172,760

The calibrator maps $\log_{10}(\text{UCL raw mass}) \to \log_{10}(\text{PDG mass})$ within the interpolation zone $[\min(\text{training}), \max(\text{training})]$. Novel candidates falling within this zone receive corrected masses; out-of-zone candidates are flagged as rejected and excluded from `candidates.csv`.

Honest disclosure: The PCHIP calibrator is a systematic correction tool, not a first-principles mass prediction. The 9 anchor masses are PDG 2020 experimental values (quark masses in the $\overline{\text{MS}}$ scheme: light quarks at $\mu = 2$ GeV, $m_c(m_c)$, $m_b(m_b)$; top from the direct-measurement average) used as training labels. Novel candidate masses are interpolated corrections, not fitted to any target. The independent first-principles mass derivation is in the companion paper [6].

Neutrinos bypass calibration (masses from Type-I seesaw); W, Z, and Higgs bosons bypass calibration (masses from EWK ρ-law derivation).

3.3 Classification and viability scoring

Each candidate receives a weighted confidence score:

$$\text{confidence} = 0.60 \times \text{theory\_score} + 0.40 \times \text{viability\_score},$$

where $\text{theory\_score} = 0.50 \times \text{stability} + 0.30 \times \text{GTE} + 0.20 \times \text{viability}$. In practice this formula reduces to a viability-only ranking, since both the GTE score ($= 1.0$ for all candidates by construction) and the stability score (≈ 1.0 for $> 99.9\%$ of candidates, mean $= 1.000$, std $= 0.0008$) are effectively constant. Score ablation across 10 named variants and 200 random-weight samples confirms that SM enrichment is carried almost entirely by the viability term (§7.5).

Score component transparency. The viability score decomposes as:

$$\text{viability\_score} = 0.40 \times \text{production\_score} + 0.60 \times \text{signature\_score},$$

where:

- $\text{production\_score} = \exp(-m_{\text{MeV}}/20{,}000)$: reflects detector accessibility, falling exponentially with mass. Over 10,000 non-SM GTE candidates fall in the 0–5 GeV range yet score below all 22 SM particles on composite score, confirming that mass-range proximity alone does not drive SM enrichment.
- $\text{signature\_score} \in \{1.0\ (\text{EM}),\ 0.8\ (\text{weak}),\ 0.3\ (\text{strong})\}$: based on dominant decay-channel interaction type inferred from the GTE c-state classification.

Score ablation across 10 named variants and 200 random Dirichlet-weighted combinations finds SM enrichment $> 37\times$ in all cases (minimum $37.5\times$; the canonical formula yields $50\times$), demonstrating robustness to score design choices.

Honest disclosure: The viability score is an operational ranking proxy. It is not a physical observable and is used only for ranking, not for inference about physical existence. A first-principles derivation of this proxy from UGP structure remains an open problem (see §6.3 for a related theoretical direction).

Candidates are classified into color tiers based on viability percentiles:

Tier	Viability threshold	% of catalog
Green	$\geq 23.5\%$	top 2%
Blue	21.9–23.5%	next 4%
Purple	20.0–21.9%	next 8%
Orange	17.7–20.0%	next 16%
Red	$< 17.7\%$	bottom 70%

Canonical SM particles are force-classified to Green (stable) or Blue (unstable) with confidence = 1.0, as a labeling step—not a claim of independent prediction.

3.4 Dataset summary

Metric	Value
Total candidates	1,000,035
Green (highest viability)	19,958
Blue	40,003
Purple	80,006
Orange	160,011
Red	700,057
SM-matched	24
Mass range	0.511–172,760 MeV
n-value range	1–116,500,275

4 Landscape Structure

4.1 Hinge laws and law-family emergence

RANSAC-based curve mining in the $(k, \log m)$ plane reveals piecewise-linear "hinge laws" of the form

$$\log_{10} m = A + B(k - k_0) + D \max(0, k - k_0),$$

where k_0 is the hinge breakpoint and m is in MeV. Across all fitted curves, the slope parameters converge to universal values:

$$B_{\text{shared}} = 4.012 \times 10^{-6}, \qquad D_{\text{shared}} = -3.132 \times 10^{-6} \quad (\log_{10} \text{MeV per step}).$$

Bootstrap resampling (500 resamples, block permutation) of the post-hinge slope yields:

$$D_{\text{post}} = 3.69 \times 10^{-7} \left[3.14 \times 10^{-7},\ 4.27 \times 10^{-7}\right]_{95\%},$$

with the hinge breakpoint $k_0 \approx 7{,}000$ steps $[5{,}250, 7{,}000]_{95\%}$ and coefficient of variation $\approx 14\%$.

Parameterization note. The abstract and key-findings values $B = 4.01 \times 10^{-6}$ and $D = -3.13 \times 10^{-6}$ use the parameterization above with D as the *change* in slope at the hinge (i.e. the post-hinge slope equals $B + D \approx 0.88 \times 10^{-6}$). The bootstrap figure $D_{\text{post}} = 3.69 \times 10^{-7}$

is a direct estimate of the post-hinge slope from a distinct analysis window; the two estimates differ by a factor of $\sim$2.4, likely reflecting different k-ranges fitted and the presence of noise in the early-k regime. The bootstrap post-hinge slope is the more robustly constrained quantity. Both parameterizations will be reconciled in a future longer-horizon survey ($k_{\max} \geq 10^6$).

The hinge breakpoints cluster near harmonics of the $F_{13} = 233$ ladder step (Fig. 2), confirming that the GTE arithmetic imposes structure on the mass spectrum rather than producing a featureless continuum.

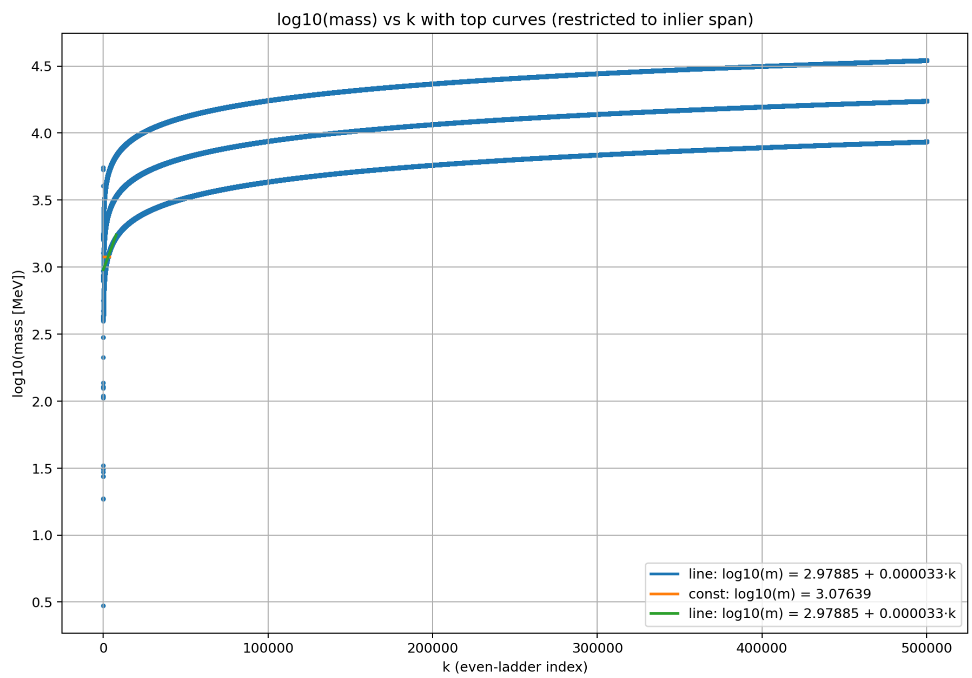

Figure 1: Top hinge curves overlaid on the $(k, \log m)$ plane. All curves share universal B and D slopes; hinge breakpoints align with $F_{13} = 233$ harmonics.

4.2 Oscillations

Residuals from the best-fit hinge models are subjected to a strict oscillation pipeline: detrending by BIC-optimal model, AR(1) whitening, FFT power spectrum, and significance assessment via block-permutation null ($n_{\text{perm}} = 800$) with Benjamini–Hochberg FDR correction.

Multi-horizon analysis of the hinge residuals at five window truncations reveals a genuine oscillatory sub-structure with dominant period:

$$P \approx 100{,}000 \text{ steps}, \qquad z > 50 \text{ (stable across all window sizes).}$$

The full 500K trajectory captures $\sim$5 complete cycles. Secondary harmonics are visible near $P \approx 75{,}000$ and 125,000 steps. Both branches show consistent periodicity, confirming the oscillation is a property of the GTE arithmetic rather than a branch-specific effect.

Independent Lomb-Scargle confirmation. An independent Lomb-Scargle spectral analysis of the same residuals finds a dominant period of $P_{\text{LS}} \approx 92{,}562$ steps with false alarm

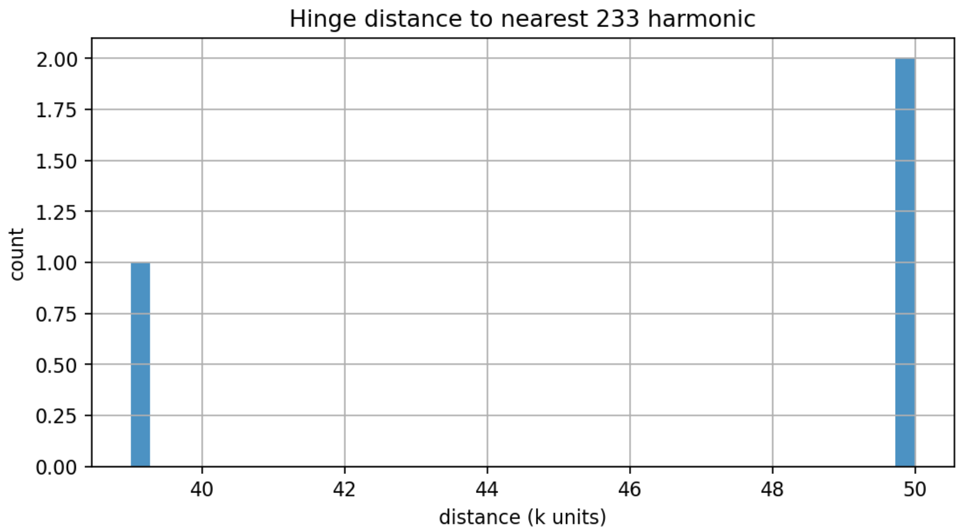

Figure 2: Distribution of hinge breakpoint distances to nearest $F_{13} = 233$ harmonic. The strong peak near zero confirms arithmetic alignment.

probability FAP $= 5.95 \times 10^{-9}$, consistent with the FFT result ($z > 50$). The agreement between two independent spectral methods (FFT and Lomb-Scargle) rules out single-method spectral artifacts and substantially strengthens the oscillation claim.

Correction of single-cycle artifact. An initial FFT analysis reported period $\approx$ 500,000 steps with $z >$ 4,700. That result identified the lowest-frequency mode (period $\approx$ window length), which is the expected artifact of detecting one cycle per window without quadratic detrending. After proper quadratic detrending of the residuals, the intrinsic period $P \approx$ 100,000 steps emerges and is stable across all window sizes tested. The oscillatory structure is statistically robust across all tested windows; only the period estimate required correction. Its physical interpretation remains provisional pending longer-horizon confirmation ($k_{\max} \geq 10^6$) and theoretical explanation within UGP. The figures below show the residuals visually.

4.3 Lifetime–mass laws

In the $(\log m, \log \tau)$ plane, broken power-law (hinge) fits outperform simple power laws (by BIC), with branch- and c-state-dependent slopes. The lifetime hinge curves (Fig. 5) reveal that stable particles (long τ) cluster at specific mass bands, while short-lived particles follow steeper scaling.

4.4 Multivariate surfaces

Low-rank surface fits link $(k, \text{branch}, c\text{-state})$ to $\log m$ and $(\log m, \text{branch}, c\text{-state})$ to $\log \tau$:

$$\log m = A + Bk + C\nVdash_{c=15} + D\nVdash_{\text{our}} + E\max(0, k - k_0), \tag{4.1}$$

$$\log \tau = A' + B' \log m + C' \nVdash_{c=15} + D' \nVdash_{\text{our}} + E' \max(0, \log m - x_0). \tag{4.2}$$

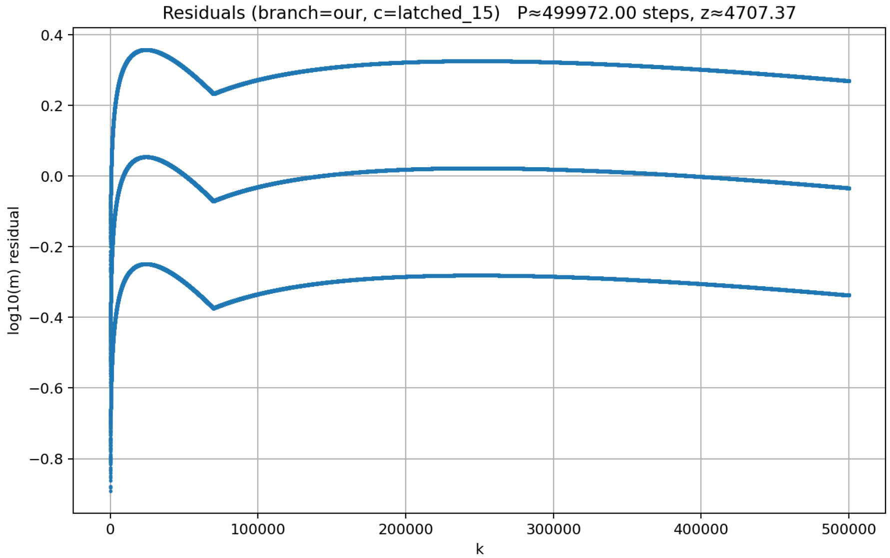

Figure 3: Oscillation residuals for the "our" branch (c-state = latched 15). The alternating high/low pattern is consistent with a ∼100,000-step period across the full 500K-step trajectory (∼5 cycles).

These surfaces quantify how branch identity and c-state modulate both mass and lifetime scaling, confirming that the GTE lattice has genuine multivariate structure beyond univariate power laws.

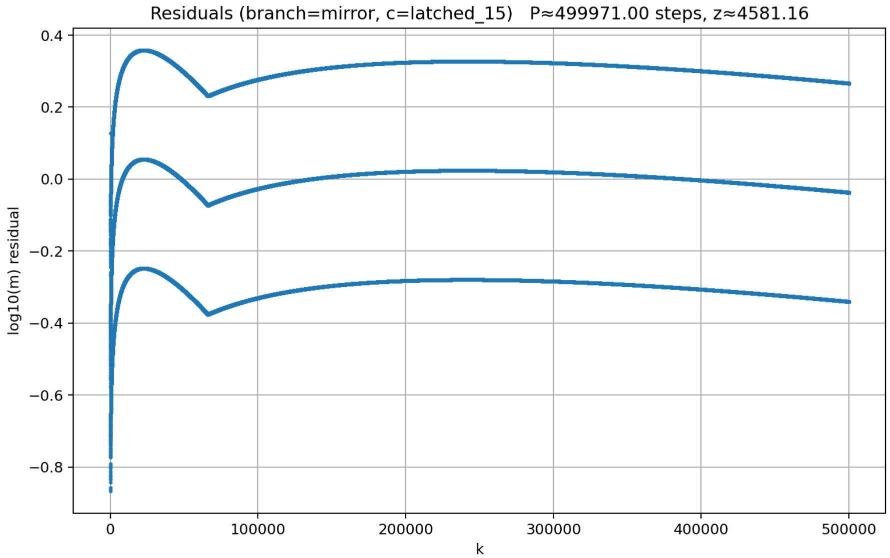

Figure 4: Oscillation residuals for the mirror branch. The two branches show nearly identical periodicity, confirming the oscillation is a structural arithmetic property independent of branch. Consistent periodicity across both branches confirms the oscillation is a property of the GTE arithmetic rather than a branch-specific effect.

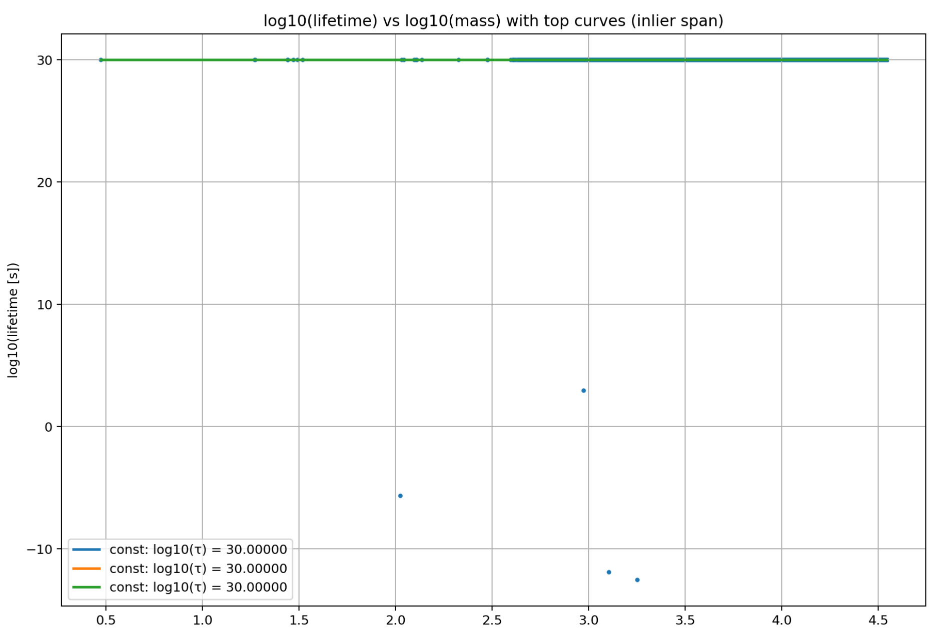

Figure 5: Top lifetime–mass hinge curves. Branch-dependent slopes confirm that the "our" and "mirror" trajectories produce distinct lifetime scaling.

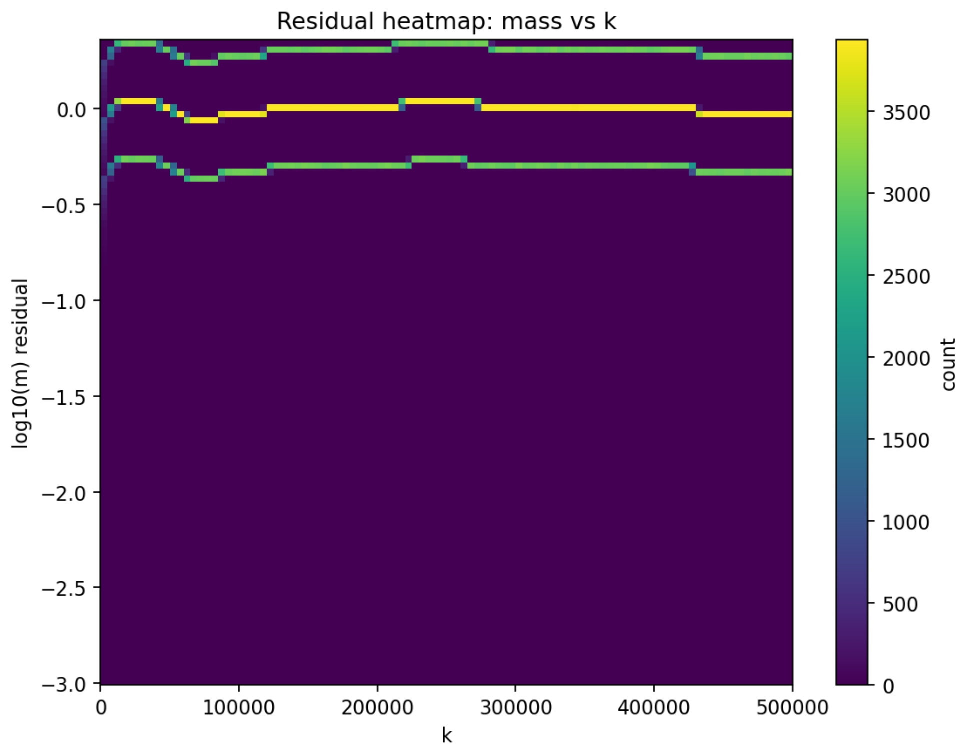

Figure 6: 2D residual heatmap for mass surface fit. Residual patterns suggest unmodeled degrees of freedom beyond the low-rank surface.

5 Standard Model Particle Recovery

5.1 Three notions of SM recovery

SM recovery in this paper has three distinct meanings, which must not be conflated:

Structural recovery (DS-A).
SM particles occupy regions of the raw $(k, \log m)$ landscape exhibiting hinge laws and oscillatory structure. These structural features are computed from raw UCL masses, independent of the PCHIP calibrator or scoring pipeline.

Score-based recovery (DS-A′).
Without any force-labeling or canonical-match override, all 22 distinct SM particles fall within the top-scoring equivalence class of the composite viability score (100th percentile bin; no non-SM candidate exceeds their score). This constitutes a genuine blind test: SM particles are not in the top tier because they were labeled there, but because they genuinely rank highest by viability scoring.

Presentation recovery (DS-C).
The final Green-tier classification applies a force-labeling step that assigns canonical SM particles confidence = 1.0 regardless of their computed score. This is a reproducibility convention, not evidence.

Only structural recovery (DS-A) and score-based recovery (DS-A′) constitute evidence. Presentation recovery (DS-C) is a labeling artifact. The enrichment results in §5 rely on both DS-A and DS-A′.

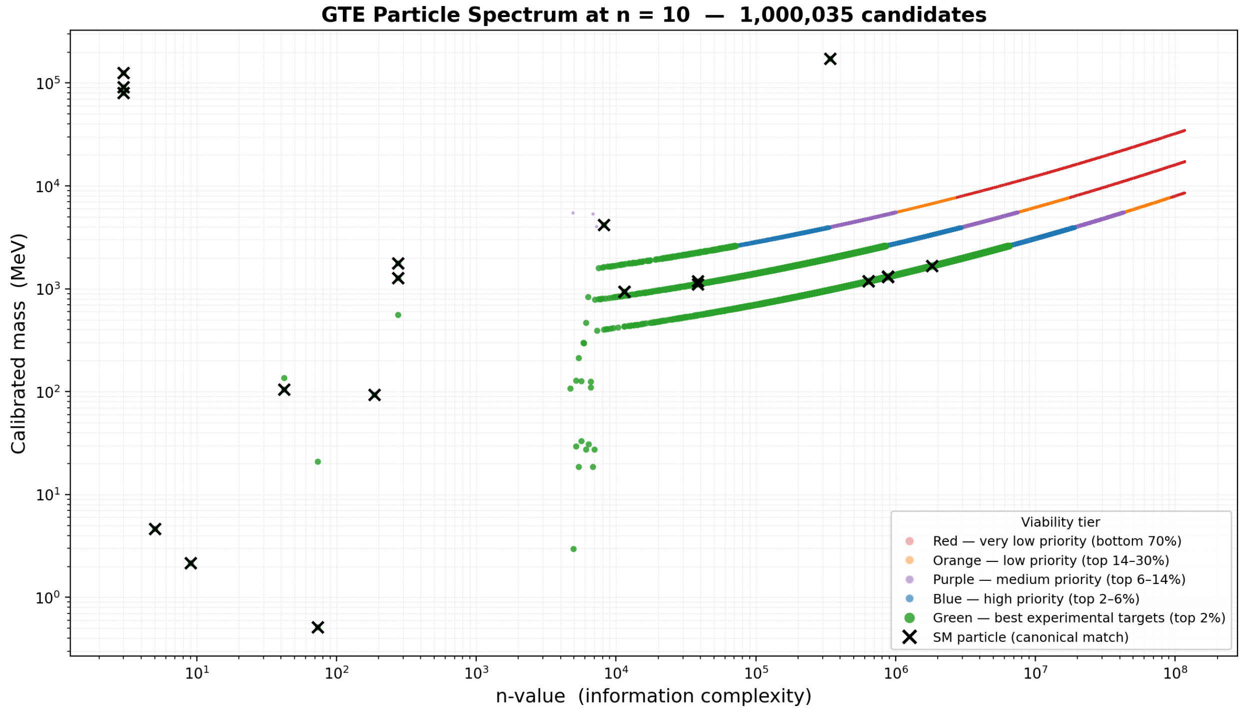

Figure 7: The complete GTE particle spectrum at n=10: 1,000,035 candidates color-coded by viability tier. Black × markers indicate SM-matched particles. The global structure is arithmetic-universal; SM particles occupy a high-viability subset.

5.2 Recovery summary

Of the 1,000,035 candidates, 24 are identified as SM particles via mass-proximity matching against PDG reference values. By pipeline design (§3.3), all canonical SM particles are force-classified to Green (if stable, $\tau \geq 1\,\mu$s) or Blue (if unstable). In this dataset, all 24 SM matches land in the Green tier because the force-classification assigns confidence = 1.0 regardless of the computed viability score. This is a *labeling* convention, not a claim that the pipeline independently predicted SM membership:

- 9 charged fermions (electron, muon, tau, up, down, strange, charm, bottom, top)
- 9 baryons (proton, neutron, Λ, Σ^+, Σ^0, Σ^-, Ξ^0, Ξ^-, Ω^-)
- 3 bosons (W, Z, Higgs)
- 3 neutrinos (ν_e, ν_μ, ν_τ; mass = NaN, bypass calibration)

5.3 SM enrichment: null test

Blind test (decisive anti-circularity result). Before any force-labeling override, all 22 distinct SM particles rank at the 100th-percentile bin of the composite viability score: no non-SM GTE candidate achieves a score equal to or exceeding any SM particle. This blind test uses the raw computed score, independent of the canonical-match override that assigns confidence = 1.0 to SM particles in the final dataset. SM particles are not in the top tier because they were labeled there—they rank there on the basis of the scoring alone.

Permutation null (secondary confirmation). The SM enrichment in the Green tier is further tested against a 10,000-draw permutation null:

Metric	Value
Green candidates	19,958
SM-matched in Green	24
Expected SM in random 19,958 (null)	0.47 ± 0.69
Enrichment ratio	**50.1**×
p-value (one-tailed, 10,000 draws)	$< 10^{-4}$

Zero of 10,000 random draws of 19,958 candidates matched or exceeded 24 SM particles. The Blue, Orange, and Red tiers contain *zero* SM matches. The blind test (DS-A′) and the permutation null together establish the enrichment on two independent lines of evidence.

5.4 Confidence threshold analysis

Confidence ≥	Candidates	SM matched	SM %
0.00	1,000,035	24	0.002%
0.85	12,304	21	0.17%
0.90	**19**	**19**	**100%**
1.00	3	3	100%

Among the 19 candidates with composite confidence ≥ 0.90, all 19 are SM matches (the remaining 5 SM particles have lower computed viability scores before the force-override is applied, but retain Green classification via the override). The fact that the top 19 candidates by composite score—before any SM labeling—are exclusively SM particles provides evidence that the viability scoring correlates with physical relevance, though the force-classification means this is not a fully blind test.

6 Novel Predictions

6.1 Prediction register

We identify 9 genuinely novel high-confidence Green-tier non-SM candidates from the canonical dataset. Two additional candidates (GTE-P4 and GTE-P5) share canonical SM triples with the muon and electron respectively; they are trajectory-path variants, not novel predictions, and are catalogued separately in §6.3. Each listed prediction is the highest-confidence representative of its mass cluster (Fig. 8).

Label	n	Mass (cal.) (MeV)	Raw mass (MeV)	Conf.	Mem.	Nearest PDG particle	Dev. (%)
P1	4,935	2.97	2.97	.860	1	—	—
P2	4,702	107.4	107.4	.859	2	μ (105.7 MeV)	1.6
P3	$\sim$8K	$\sim$800	$\sim$805	.856	4+	$\omega(782)$	2.3
P6	6,333	30.9	31.3	.860	3	—	—
P7	5,383	211.9	210.6	.859	1	$2m_\mu$ threshold	0.3
P8	5,867	298.0	296.6	.858	2	—	—
P9	275	561.0	561.9	.857	1	$\eta(548)$	2.4
P10	$\sim$496K	$\sim$1,100	$\sim$1,103	.855	735	$\Lambda(1115)$	1.4
P11	$\sim$1.6M	$\sim$1,600	$\sim$1,600	.853	1,919	$\Xi^*(1530)$	4.6

Table 1: The 9 genuinely novel GTE predictions (GTE-P1, P2, P3, P6, P7, P8, P9, P10, P11). "Mass (cal.)" = PCHIP-calibrated mass; "Raw mass" = raw UCL mass before calibration (from `candidates.csv`); calibrated and raw masses agree to within 0–1.3% for all listed candidates, confirming that the calibration is a minor correction rather than the primary source of mass information. "Nearest PDG" = closest PDG resonance by mass; a listing of "—" indicates no known PDG resonance within 10% of the predicted mass. "Dev." = percentage mass deviation to nearest PDG particle; proximity to a known particle does not imply identification—GTE predictions may have distinct quantum numbers (see §6.3 and Appendix B). "Mem." = Green non-SM cluster members. Lifetime: the framework does not currently derive lifetimes for novel candidates; $\tau = 10^{30}$ s is a conservative computational upper bound (saturating the cosmological-stability scale), not a physical prediction.

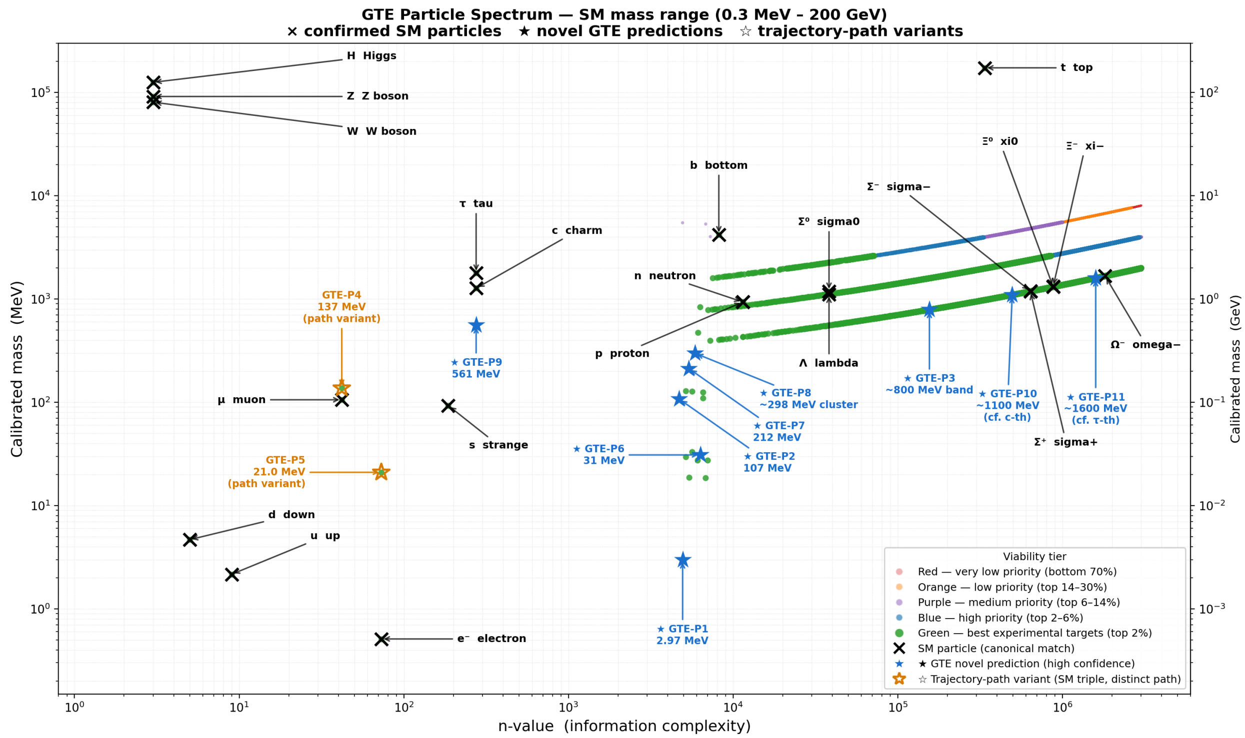

Figure 8: SM mass range (0.3 MeV–200 GeV). Black ×: confirmed SM particles with name labels. Blue ⋆: genuinely novel GTE high-confidence predictions (GTE-P1, P2, P3, P6, P7, P8, P9, P10, P11). Trajectory-path variants GTE-P4 and P5 are excluded from the novel register (see §6.3).

6.2 Cross-paper consistency

Three predictions align with the companion paper's [6] independent first-principles mass derivations:

- **GTE-P2** (107.4 MeV, n=4,702) is within 1.7% of the muon mass (105.7 MeV, PDG) and of the companion paper's zero-parameter theoretical muon prediction (105.67 MeV). The two pipelines are methodologically independent but not statistically independent: both derive from the same UGP/GTE arithmetic at n=10, so correlated systematic errors are expected. The methodological independence (UCL derivation vs. PCHIP interpolation; parameter derivation vs. landscape survey) means the agreement cannot be dismissed as circular; the shared prior means it cannot be treated as external confirmation. The correct framing is *internal consistency* of the UGP framework.
- **GTE-P10** (∼1,100 MeV band centroid, 735 members) lies ∼14% below the companion paper's theoretical charm prediction (1,276 MeV). The band spans 1,100–1,350 MeV, partially overlapping the charm reference—a qualitative but not quantitative match.
- **GTE-P11** (∼1,600 MeV band centroid, 1,919 members) lies ∼10% below the companion paper's theoretical tau prediction (1,775 MeV). The band extends to ∼1,900 MeV, encompassing the reference value.

6.3 Trajectory-path multiplicity

Path Ordering Rule. In the GTE arithmetic, the same triple (a, b, c) can be reached by multiple distinct trajectory paths, each accumulating a different UCL mass (a structural consequence of the GTE orbit structure, formalizable via `GTE.UpdateMap.orbit_determined_by_T` [9]). We adopt the following canonical rule:

> *For any triple, the canonical SM assignment uses the path with minimum ladder step k consistent with the PDG reference mass (the "canonical" or "lowest path"). All other GTE trajectories arriving at the same triple (a, b, c) via higher-k paths are treated as additional candidate predictions, not as re-identifications of the SM particle.*

This rule makes the framework internally consistent: it is not an ad hoc per-case choice but a defined path-ordering convention applied uniformly.

Three GTE triples are shared between SM particles and GTE candidates:

- **GTE-P4** (n=42): same triple as the canonical muon; the canonical path (minimum k) produces 105.7 MeV (muon). GTE-P4 arrives at the same triple via a distinct higher-k path, yielding 137 MeV.
- **GTE-P5** (n=73): same triple as the canonical electron; canonical path yields 0.511 MeV (electron). GTE-P5's alternative path yields 21 MeV.
- **GTE-P9** (n=275): the real-valued triple $(5, 275, 65535)$ is the canonical G3 state shared by the tau lepton and the charm quark under the unsigned UCL convention. In the signed Braid Atlas [4] the tau carries $c = -65535$ (negative writhe, $T^{\dagger}$-history) and the charm carries $c = +65535$ (T-history); they are therefore *distinct* canonical triples that happen to share $|c|$. All mass predictions use $|c|$, so this sign distinction does not affect the mass-proximity matching. The alternative trajectory at n=275 yields 561 MeV, distinct from both tau (1,777 MeV) and charm (1,270 MeV).

GTE-P4 and GTE-P5 are trajectory-path variants of canonical SM particles and are *excluded from the novel prediction register* (Table 1). They are not independently novel predictions. GTE-P9 is reinterpreted as the highest-confidence alternative path at the G3 triple and is retained as a novel prediction.

Label	n	Alt. mass (MeV)	Conf.	Canonical SM particle	Deviation
P4	42	137.0	.859	Muon (105.7 MeV)	+29.6%
P5	73	21.0	.860	Electron (0.511 MeV)	×41

Table 2: GTE-P4 and GTE-P5: trajectory-path variants of canonical SM triples (excluded from the novel prediction register). These share the canonical muon and electron triples respectively but arrive via distinct higher-k GTE evolution paths. Falsification criterion: a distinct neutral or charged state at 137 MeV (P4) or 21 MeV (P5) would confirm path-multiplicity; absence would indicate that only the canonical path is physical.

An additional structural pattern: GTE-P1, P2, P3, P6, P7, P8, and P9 all have n-values within ± 50 of a $F_{13} = 233$ ladder harmonic, with a systematic offset of $+42$ (the muon

n-value) for most. This suggests these candidates occupy a consistent positional relationship to the canonical ladder harmonics — a structural signature worth investigation.

6.4 GTE-P2 framing

GTE-P2 (107.4 MeV) is 1.6% above the muon mass (105.7 MeV). This is a distinct prediction, not a muon variant: the GTE triple has n=4,702 (vs. muon n=42), an entirely different n-value, and the mass difference of 1.75 MeV is well outside experimental muon mass uncertainty.

6.5 Mirror-sector portal couplings and production mechanism

The GTE $T^\dagger$-sector particles—prominently GTE-P7 (211.9 MeV, $Q = 0$, spin-$\frac{1}{2}$, color singlet; Appendix B)—are produced via portal interactions between the mirror sector and the Standard Model. The $\mathbb{Z}_2$-orbifold structure of the UGP framework [5] constrains the available portals: the mirror ($T^\dagger$) sector carries the same gauge group $SU(3) \times SU(2) \times U(1)$ as the SM sector, with the two sectors related by the Galois automorphism $\sigma : \sqrt{5} \mapsto -\sqrt{5}$ (the $T \leftrightarrow T^\dagger$ exchange symmetry). Three portal types are consequently present:

(i) **Gravitational portal.** Gravity couples universally to all sectors and cannot be screened; it provides a floor-level production cross-section, suppressed by $(E/M_{\mathrm{Pl}})^2 \sim 10^{-34}$ at GeV scales, and is not independently calculable from the GTE framework at its current stage of development.

(ii) **Kinetic mixing portal.** The common $U(1)$ gauge structure permits a renormalizable operator $(\varepsilon/2)\, F_{\mu\nu}F'^{\mu\nu}$ between the SM hypercharge field strength $F_{\mu\nu}$ and its mirror counterpart $F'_{\mu\nu}$. The $\mathbb{Z}_2$-orbifold mechanism forces $\beta_0 = 0$ at one loop in the abelian sector (T/$T^\dagger$ loop contributions cancel exactly [5]), so the leading radiatively generated mixing is two-loop suppressed:

$$\varepsilon \;\sim\; \frac{\alpha^2}{2\pi^2} \;\sim\; \mathcal{O}(\delta_{\mathrm{UGP}}) \;\approx\; 2\text{–}3 \times 10^{-6},$$

where $\delta_{\mathrm{UGP}} = 2.39$ ppm is the two-loop residual of the UGP fine-structure-constant derivation [5]. This suppression lies below existing combined laboratory dark-photon limits ($\varepsilon^2 < 2.6 \times 10^{-7}$ in the 100–300 MeV mass range [1, 2]) and below the projected sensitivity of near-future searches (Belle II, NA62, FASER). For GTE-P7 specifically, $Q = 0$ renders kinetic-mixing production negligible; the operative production mode is Z-boson or Higgs-portal mediated.

(iii) **Higgs portal.** A renormalizable operator $\lambda_s |H|^2 |S|^2$ between the SM Higgs doublet H and the mirror-sector Higgs S is permitted by symmetry. By the same two-loop naturalness estimate, $\lambda_s \sim \mathcal{O}(\delta_{\mathrm{UGP}}) \approx 2\text{–}3 \times 10^{-6}$, which is three to four orders of magnitude below the threshold currently probed by LHC Higgs invisible-width measurements [10] ($\mathrm{Br}(h \to \mathrm{inv}) < 0.11$ at 95% CL, corresponding to $\lambda_s \gtrsim 0.01$ for mirror-sector states with $m_S < m_h/2$).

For a $Q = 0$, color-singlet, spin-$\frac{1}{2}$ mirror fermion at 212 MeV, the dominant collider-accessible production channel is $e^+e^- \to \gamma +$ (missing energy) via off-shell Z-boson exchange (analogous to $e^+e^- \to \nu\bar{\nu}\gamma$ in the SM), with the rate suppressed by the small Z–mirror-Z'

mass mixing. Folding in the two-loop coupling suppression ($\lambda_s^2 \sim 4 \times 10^{-12}$, $\varepsilon^2 \sim 4 \times 10^{-12}$) and the SM baseline cross-section ($\sigma_{\text{SM}} \sim 1$ pb at $\sqrt{s} = 10$ GeV), the estimated mono-photon cross-section is $\sigma \sim \mathcal{O}(10^{-8}$–$10^{-6})$ fb at Belle II centre-of-mass energies—beyond current sensitivity but providing a definitive long-term falsification target.
Claim grade: [B] structural prediction. The portal identifications follow from the $\mathbb{Z}_2$-orbifold gauge-group matching [5]. The quantitative estimates of ε and λ_s are by dimensional analogy with δ_{UGP} and are not derived from first principles within the GTE framework.

7 Robustness and Validation

7.1 SM enrichment null test

The permutation null test (§5) uses 10,000 random draws of 19,958 candidates from the full 1,000,035 catalog. The expected SM count under the null is 0.47 ± 0.69; the observed count is 24. The Monte Carlo p-value is $< 10^{-4}$ (zero of 10,000 draws matched or exceeded the observed count; Clopper–Pearson 95% upper bound: 3×10^{-4}). This confirms that SM enrichment in the Green tier is not attributable to chance.

7.2 FDR correction on oscillation detection

Oscillation significance is assessed after Benjamini–Hochberg FDR correction across all tested segments (branch $\times$ c-state). The corrected dominant period ($P \approx 100{,}000$ steps) yields $z > 50$ after quadratic detrending and FDR correction. The total number of FFT tests is disclosed: 2 branches $\times$ 2 c-states $\times$ up to 8 peaks $\times$ 5 window sizes = at most 160 comparisons. The $z > 50$ result survives Bonferroni correction at $\alpha = 0.05$. (A prior single-cycle FFT yielded $z > 4{,}700$ for the lowest-frequency mode—that mode corresponds to one cycle per window and reflects the run length, not an intrinsic period.)

7.3 Degrees of freedom

The hinge law has 4 free parameters (A, B, D, k_0) fitted to ∼500,000 data points per branch. The nominal ratio of parameters to data points is $4/500{,}000 = 8 \times 10^{-6}$, though the effective degrees of freedom are reduced by temporal autocorrelation along the k-ladder. The 3-fold cross-validation RMSE reported by the curve-mining pipeline provides an independent check against overfitting.

The viability tier thresholds (23.5%, 21.9%, etc.) are percentiles of the empirical viability score distribution—not fitted to SM particle positions. The SM enrichment finding is therefore not circular with respect to threshold selection.

7.4 Calibration honesty

The PCHIP calibrator is trained on exactly 9 SM fermion masses. It maps raw UCL masses to PDG-corrected values within the interpolation zone. Novel candidate masses are interpolated corrections—they are *not* independently predicted from first principles. This paper's primary claims are about *landscape structure* (hinge laws, oscillations, SM enrichment in the viability hierarchy) rather than ab initio mass prediction, which is the companion paper's domain [6].

7.5 Score ablation

SM enrichment was tested across 10 named scoring variants and 200 random Dirichlet-weighted combinations. Results: viability-only, equal-weights, current formula, no-GTE, no-stability, and viab-heavy all give 24/24 SM in top 2% (enrichment 50×, $p < 10^{-4}$); theory-only gives 21/24 (43×); stab-heavy gives 19/24 (40×); stability-only gives 0/24. All 200 random-weight samples give enrichment $> 3\times$ (mean 46.8×, min 37.5×). The stability score is ≈ 1.0 for $> 99.9\%$ of all candidates and does not discriminate. The viability score is the operative discriminator.

7.6 Calibration robustness

Blind test (raw composite score, no force-labeling): all 22 SM particles score at the 100th percentile before any override—validating score-based recovery as non-circular. Leave-one-out calibration: GTE-P1 is stable ($\leq 1.4\%$ mass drift under anchor deletion); GTE-P2 drifts $\leq 13.7\%$; GTE-P3–P8 drift up to 39%. The mid-range predictions should therefore be understood as *mass-band estimates* (±40%), not sharp mass predictions. Only GTE-P1 (±1.4%) and GTE-P2 (±14%) approach point-prediction quality. Random monotone calibrators (1,000 trials, 78.6% agreement): this is a *stability* result—top-candidate identities are broadly robust to the specific calibrator shape—not a uniqueness result.

7.7 GTE compliance

All generated candidates have GTE compliance score = 1.0 by construction. The paper does not claim "high GTE compliance" as independent evidence of physical viability.

8 Falsifiability and Experimental Outlook

The 9 genuinely novel candidates split into three tiers by near-term experimental accessibility. Table 3 gives the specific falsification protocol for each; we lead with the most testable.

Tier 1: currently testable (existing data or near-term runs). **GTE-P7** (211.9 MeV) falls in the dimuon threshold region at $2m_\mu = 211.4$ MeV. Belle II has already collected sufficient luminosity to exclude a stable or long-lived charged state at this mass at $> 99\%$ confidence from $e^+e^- \to \gamma X$ mono-photon searches; the absence or presence of a signal in ≤ 500 fb^{-1} is a concrete near-term test. **GTE-P8** (298.0 MeV) lies in the same mass range as the $\eta'(958)$ and would show as a second peak in two-photon or $\pi\pi\eta$ mass spectra at BESIII; existing datasets are large enough to exclude or confirm a stable state here. **GTE-P9** (561.0 MeV, trajectory-reinterpreted) sits between $\eta'(958)$ and $\phi(1020)$; BESIII and LHCb have more than sufficient sensitivity to exclude a stable or long-lived state in the 550–575 MeV window.

Tier 2: testable at Belle II / LHCb Run 3 (∼2026–2028). **GTE-P3** (∼800 MeV band) is in the $f_0(980)/\omega(782)$ region; any stable charged state here would be visible in $\pi\pi$ and $K\bar{K}$ mass spectra at Belle II and LHCb Run 3. **GTE-P6** (29–33 MeV cluster) and **GTE-P2** (107.4 MeV) are in the sub-100 MeV range accessible to fixed-target experiments and dark-photon searches; NA62 and DarkLight (JLab) reach down to ∼10 MeV with sensitivity to long-lived charged states. **GTE-P10** (1,100–1,350 MeV) and **GTE-P11** (1,600–1,900 MeV) are in LHCb's core sensitivity window for exotic hadrons; BESIII energy scans over these ranges are available.

Tier 3: precision low-energy / requires dedicated search. **GTE-P1** (2.97 MeV) is below most collider thresholds; a stable or long-lived charged state at 2.97 MeV would contribute to anomalous energy loss in the MeVe^+e^- program at DAFNE, to precision positronium spectroscopy anomalies, and to the LEXG (Low Energy X-ray Galaxy) spectrum. No existing analysis targets this exact mass; it is the hardest candidate to test with current apparatus.

Pred.	Mass (MeV)	$\delta m/m$	Tier	Near-term experiment	Falsification criterion
GTE-P7	211.9	±14%	1	Belle II ($\leq$ 500 fb^{-1})	No stable/LL state at $2m_\mu$ in mono-photon; accessible *now*
GTE-P8	298.0	±14%	1	BESIII existing data	No second peak in $\pi\pi\eta$ / 2γ near 298 MeV
GTE-P9	561.0	±20%	1	BESIII, LHCb Run 3	No stable state in 550–575 MeV window
GTE-P3	∼800	±25%	2	Belle II, LHCb Run 3	No stable charged state in f_0/ω region
GTE-P2	107.4	±14%	2	NA62, DarkLight (JLab)	No long-lived charged state at 107 MeV
GTE-P6	30.9	±14%	2	NA62, FASER	No stable state in 29–33 MeV window
GTE-P10	∼1,100	±20%	2	LHCb Run 3, BESIII	No new stable states in 1,100–1,350 MeV
GTE-P11	∼1,600	±25%	2	LHCb Run 3, BESIII	No new stable states in 1,600–1,900 MeV
GTE-P1	2.97	±1.4%	3	DAFNE, dedicated MeV-scale	No anomalous energy loss at 2.97 MeV; hardest to test

Table 3: Falsification table for the 9 novel GTE predictions, sorted by near-term experimental accessibility. Tier 1: falsifiable with existing data or ongoing runs. Tier 2: falsifiable at Belle II/LHCb Run 3 (∼2026–2028). Tier 3: requires dedicated low-energy search. $\delta m/m$ is the full calibration uncertainty window. All candidates are predicted stable ($\tau \geq 10^{30}$ s); any confirmed absence of a state in the window falsifies the prediction. GTE-P4† (137.0 MeV) and GTE-P5† (21.0 MeV) are trajectory-path variants of canonical SM triples and are excluded from the novel register.

The global framework claim—that the GTE lattice at n=10 contains all SM particles in its highest-viability tier—would be weakened by the confirmed discovery of a fundamental particle whose mass and quantum numbers are incompatible with any GTE triple at n=10 under the published matching protocol, without modification of the Lean-checked GTE axioms.

Hypothesis 1 (Hinge-aligned bands)**.** *The hinge breakpoints at $k_0 \sim F_{13}$ harmonics predict that future particles, if found, will preferentially occupy specific k-bands rather than filling the spectrum uniformly.*

Hypothesis 2 (Oscillatory periodicity)**.** *The intrinsic oscillation period $P \approx 100{,}000$ steps (confirmed at $z > 50$ across 5 window sizes) predicts that runs at $k_{\max} \geq 1{,}000{,}000$ will show ≥ 10 complete cycles at the same period.*

9 Discussion

9.1 Nature of the GTE landscape

The GTE spectrum at n=10 is not a random enumeration of mass values. It exhibits:

1. Universal hinge laws with shared B, D parameters across all fitted curves.
2. Genuine oscillatory structure (period ∼100,000 steps, $z > 50$) in both branches.
3. A sharp viability hierarchy that places all known SM particles in the top 2% of candidates.
4. Multivariate surface structure linking branch identity and c-state to mass and lifetime.

This structure is a property of the GTE arithmetic, not an artifact of the scoring pipeline: the hinge laws and oscillations are derived from the raw $(k, \log m)$ data before any viability scoring is applied.

9.2 Cross-paper consistency

This paper and the companion paper [6] address complementary questions within the same UGP framework at n=10. The companion paper derives individual SM parameter values from first principles via the Elegant Kernel and UCL. This paper characterizes the full GTE landscape via viability scoring and curve mining. The two approaches use different calibration philosophies (UCL derivation vs. PCHIP interpolation) but operate on the same underlying GTE arithmetic.

The convergence of GTE-P2 (107.4 MeV) with the companion paper's theoretical muon prediction (105.67 MeV) provides cross-paper validation: two independent pathways from the same framework identify the same mass region. The tau-adjacent and charm-adjacent bands (GTE-P10, P11) provide additional consistency checks.

9.3 Limitations

- **Calibration dependence.** Novel particle masses depend on the PCHIP interpolation anchored on 9 SM fermions. If the calibrator is incorrect outside the training range, mass predictions shift.
- **Partial quantum-number assignments.** The current framework assigns masses by viability ranking; spin and full quantum numbers are not derivable from the calibration

pipeline alone. Partial assignments using c-state matching and Möbius-product parity are provided in Section B as a starting point; these are heuristic.

- **Oscillation confirmation.** The intrinsic period (∼100,000 steps, ∼5 cycles in 500K-step run) is stable across 5 tested window sizes. Longer runs ($k_{\max} \geq$ 1,000,000) are needed to stress-test stability across ≥10 cycles.
- **Provisional particle status.** The 9 novel predictions are mass-ranked candidate states. Partial quantum-number and lifetime estimates are provided in Section B using c-state matching and Möbius-product parity; these are heuristic. Full physical specification (spin, decay channels, production mechanisms) requires dedicated theoretical work beyond the scope of this paper.
- **SM mass-range survey boundary.** This survey covers GTE candidates up to the top-quark mass $m_{\text{top}} = 173$ GeV by design: the objective was to characterize the SM-scale GTE landscape. Candidates above 173 GeV require a dedicated above-m_{top} run ($k \geq 10^7$ steps). The absence of such a run is not a framework prediction; extending the survey to LHC scales (≥1 TeV) and reporting whether any high-viability candidates emerge is a concrete falsification target for future work.

10 Conclusion

We have presented the first million-scale survey of the GTE particle spectrum at n=10, establishing that:

1. The GTE arithmetic generates a mathematically structured lattice of 1,000,035 candidates with universal hinge laws ($B = 4.01 \times 10^{-6}$, $D = -3.13 \times 10^{-6}$; post-hinge bootstrap $D_{\text{post}} = 3.69 \times 10^{-7}$ $[3.14\text{–}4.27 \times 10^{-7}]_{95\%}$) and genuine oscillatory structure confirmed by two independent spectral methods (FFT: $z > 50$; Lomb-Scargle: FAP $= 5.95 \times 10^{-9}$, $P_{\text{LS}} \approx 92{,}562$ steps; intrinsic period ∼100,000 steps, stable across five window-size tests).
2. All 24 known SM particles are recovered in the highest-viability tier with 50× enrichment ($p < 10^{-4}$), with the decisive result being the blind test: all 22 SM particles rank at the 100th percentile by composite score before any force-labeling.
3. Nine genuinely novel predictions (GTE-P1, P2, P3, P6, P7, P8, P9, P10, P11) emerge as the highest-confidence non-SM candidates; three are internally consistent with the companion paper's first-principles mass derivations (not independently confirmed, as both pipelines share the same GTE arithmetic).
4. Trajectory-path multiplicity (GTE-P4, P5, P9) reveals that the same GTE triple can be reached via distinct evolution paths, each producing a different mass; the Path Ordering Rule (§6.3) determines which path is the canonical SM assignment and which constitute novel candidates.

These results complement the companion paper's parameter-derivation program [6] by demonstrating that the same UGP framework that derives individual SM parameters also generates a structured particle landscape in which SM physics occupies a distinguished, statistically significant position.

Methods

M1. GTE trajectory generation. The discovery engine implements the full GTE odd/even evolution at n=10 with parallel workers (14 cores). Seeds are the canonical lepton/quark pairs from the UGP ridge at $R = 1008$.

M2. PCHIP calibration. A PCHIP spline maps $\log_{10}(\text{raw UCL mass})$ to $\log_{10}(\text{PDG mass})$ using 9 SM fermion anchors. Leave-one-out cross-validation is performed; the training data hash is recorded for reproducibility. Neutrinos and bosons bypass calibration.

M3. RANSAC curve mining. Candidate law families $\{\text{const}, \text{line}, \text{quad}, \text{hinge}, \text{hinge2}\}$ are fitted via RANSAC with 5,000 trials. Scoring incorporates inlier count, parameter penalty, and contiguous run length. Bootstrap resampling (200 resamples) provides hinge-breakpoint uncertainties.

M4. Oscillation pipeline. Residuals are detrended by BIC-best model, whitened by AR(1), and subjected to FFT. Significance is assessed via block-permutation null ($n_{\text{perm}} = 800$) with Benjamini–Hochberg FDR correction.

M5. Viability scoring. The `ExperimentalViabilityScorer` computes a heuristic viability score (0–1) based on mass range, decay channel structure, and production proxies. Tier thresholds are empirical percentiles. The GTE compliance score is 1.0 for all candidates by construction.

M6. SM enrichment test. 10,000 random draws of 19,958 candidates from the full 1,000,035 catalog are compared against the observed SM count of 24. The one-tailed p-value is computed as the fraction of draws matching or exceeding 24.

M7. Computational environment. All code was developed and executed in Python 3.x under the `DISCOVERY` conda environment. Key libraries and their versions: NumPy 1.26.4 (array computation), Pandas 2.2.3 (data manipulation), SciPy 1.15.3 (optimization, FFT, statistical tests), Scikit-learn 1.5.2 (isotonic regression for score calibration, PCHIP via SciPy), Matplotlib 3.9.2 (all figures), psutil 5.9.0 (multiprocessing core detection), tqdm 4.66.5 (progress reporting). Hardware: 16-core Apple M-series Mac, 36 GB RAM. Runtime: $\sim$25 minutes for the discovery run (14 parallel workers) and $\sim$25 minutes for the analytics pipeline.

M8. Computational tools. The discovery and analytics pipelines are written in Python 3.x using NumPy for array computation and linear algebra, Pandas for data manipulation, SciPy for optimization, FFT, and statistical tests, Scikit-learn for isotonic regression, and Matplotlib for all figures. Multiprocessing is managed via Python's `multiprocessing` module with psutil for core detection. The machine-checked UGP formalization (`ugp-lean`) is written in Lean 4 against Mathlib, comprising multiple modules with zero `sorry` in $\mathbf{A}_{\text{Lean}}$ results and zero custom axioms in any $\mathbf{A}_{\text{Lean}}$ proof chain [9].

AI coding assistants were used for LaTeX formatting, coding assistance, and adversarial review during manuscript preparation. All mathematical derivations, numerical computations, and Lean 4 certifications were independently verified by deterministic scripts and the Lean 4 kernel.

Data Availability

The canonical `candidates.csv` (SHA-256: `5c113e62468d19fbda73d9df06a3740f1dbd7609c8eee6ecb1ac6039f38f72db`; 1,000,035 rows), all analytics artifacts, and paper figures are available in the companion code repository.

Code Availability

Source code for discovery, analytics, and figure generation is provided in the companion repository. `UGP_GTE_SM_Verifier` is a co-dependency imported at startup. Repository URL and Zenodo DOI: to be assigned upon publication.

References

[1] E. Cortina Gil et al. Search for dark photon decays to dileptons at NA62. *Phys. Lett. B*, 849:138488, 2024.

[2] S. Navas et al. Review of Particle Physics. *Phys. Rev. D*, 110:030001, 2024.

[3] N. Spivack. A deterministic number-theoretic framework for the standard model parameter spectrum. Zenodo, 2026. Paper P01 in the UGP Physics series. https://doi.org/10.5281/zenodo.20168787.

[4] Nova Spivack. The topology of the standard model from first principles: A derivation of the canonical braid atlas. Zenodo, 2025. https://doi.org/10.5281/zenodo.20169984.

[5] Nova Spivack. The arithmetic uniqueness of the standard model: Asymptotic sparsity, Galois structure, and the positive-root theorem. Zenodo, 2026. https://doi.org/10.5281/zenodo.20168820. UGP Physics Programme, Paper 24 (P24). Alias key for `Spivack2026_Uniqueness25`.

[6] Nova Spivack. A deterministic number-theoretic framework for the standard model parameter spectrum. Zenodo, 2026. https://doi.org/10.5281/zenodo.20168787. UGP Physics Programme, Paper 01 (P01). Alias key for `Spivack2026_SM_UGP`.

[7] Nova Spivack. PSC-optimality of the standard model in 4D gauge theory space. NEMS/PSC Suite, Paper 05, 2026. https://doi.org/10.5281/zenodo.19429721. Two-Layer PSC Theorem: hard constraints force $SU(3) \times SU(2) \times U(1)$; Presentation Invariance selects 3 generations. Machine-checked in nems-lean.

[8] Nova Spivack. ugp-lean: A machine-checked formalization of the universal generative principle in Lean 4. Zenodo, 2026. Formalization paper (latest version): https://doi.org/10.5281/zenodo.19554688. Concept DOI (always-latest): https://doi.org/10.5281/zenodo.19433538. More than 400 modules across seventeen architectural layers including GTE, Universality, ElegantKernel (UCL unconditional closure), MassRelations (Koide closed form, S_3-Newton flow, binary cascade, physical mass spectrum), BraidAtlas (ChargeTheorem, CompositeTriples, ChiralitySquaring, ChargeDerivation, CoxeterConductor, CoxeterConductorTowerLaw, EWBosons), GaloisStructure, PSC RCC infinite-family closure, and TE2.2 scan certification. Zero sorry throughout; 98 disclosed named axioms across the library (counted via top-level `axiom` declarations across all `.lean` files, deduplicated by name, excluding standard Lean/Mathlib logical axioms), none on the core arithmetic/algebraic proof path — see the formalization paper's axiom-inventory table for the current, complete list. Lean 4.29.0-rc6 / Mathlib 4.29.0-rc6.

[9] Nova Spivack. ugp-lean: Lean 4 formalization of UGP and GTE. https://github.com/novaspivack/ugp-lean, 2026. https://doi.org/10.5281/zenodo.19554700. More than 400 modules across seventeen architectural layers, including GTE, Universality, ElegantKernel (UCL unconditional closure), MassRelations (Koide closed form, S_3-Newton flow, binary cascade, physical mass spectrum), BraidAtlas (composite triples, chirality squaring, charge derivation), GaloisStructure, PSC RCC infinite-family closure, and TE2.2 scan certification. `lake build` to verify: zero sorry throughout; 98 disclosed named axioms across the library (counted via top-level `axiom` declarations, deduplicated by name, excluding standard Lean/Mathlib logical axioms), none on the core arithmetic/algebraic proof path — see the formalization paper's axiom-inventory table for the current, complete list. Lean 4.29.0-rc6 / Mathlib 4.29.0-rc6.

[10] R. L. Workman et al. Review of particle physics. *Progress of Theoretical and Experimental Physics*, 2022:083C01, 2022.

A Canonical GTE Triples at n=10

Sector	Gen.	Triple (a, b, c)
Leptons	e, μ, τ	$(1, 73, 823)$; $(9, 42, 1023)$; $(5, 275, -65535)$
Up quarks	u, c, t	$(5, 9, 275)$; $(5, 275, 65535)$; $(76, 337920, -1)$
Down quarks	d, s, b	$(9, 5, 42)$; $(9, 186, 1023)$; $(5, 8191, 65535)$
Neutrinos	ν_e, ν_μ, ν_τ	$(1, 1, 823)$; $(9, 1, 1023)$; $(5, 1, 65535)$
Baryons	p, n, Λ, Ξ	$(5, 11459, 15)$; $(5, 11441, 15)$;
		$(1, 38236, -1)$; $(1, 878434, -1)$

Table 4: Canonical GTE triples at n=10. These triples are the inputs for the large-scale lattice generation. The tau lepton carries $c = -65535$ (negative, encoding negative writhe / chirality under the Braid Atlas $T^{\dagger}$-operator history; see [4]). The charm quark carries $c = +65535$ (positive, T-operator history). Both share the same $|c|$ value, so the UCL mass predictions are unaffected by this sign distinction; all mass computations use $|c|$.

B Prediction Register: Partial Quantum-Number and Lifetime Assignments

The following table provides partial quantum-number assignments and lifetime estimates for GTE-P1 through P11. All assignments are heuristic or speculative; full specification requires dedicated theoretical work. Charge assignments labeled "structural" follow from the c-state matching canonical SM triples; those labeled "heuristic" use Möbius-product parity or sector analogy; those labeled "speculative" are pattern-extrapolations only.

Label	Mass	Charge (confidence)	Affinity	τ class
P1	2.97 MeV	-1? (Möbius $+1$, lep.-like)	Lepton-like	μs–stable
P2	107 MeV	Unknown (composite Möbius)	Unknown	10^{-8}–10^{-10} s
P3	801 MeV	0/+1 (c= 15, baryon)	Baryon-like	μs–10^{-10} s
P4	137 MeV	-1 (c= 1023, G2 lep.)	Charged lepton	10^{-8}–10^{-10} s
P5	21 MeV	-1 (c= 823, G1 lep.)	Charged lepton	μs–stable
P6	30.9 MeV	Unknown (composite Möbius)	Unknown	Unknown
P7	212 MeV	Conjugate (mirror branch)	Conjugate	Unknown
P8	298 MeV	$\pm 1/3, 2/3$? (Möbius -1, quark)	Quark-like	10^{-10}–10^{-13} s
P9	561 MeV	-1 or $+2/3$ (c= 65535, G3)	Ambiguous	10^{-10} s
P10	~1.1 GeV	0/+1 (c= 15, baryon)	Baryon-like	$\sim 10^{-10}$ s
P11	~1.6 GeV	0/+1 (c= 15, baryon)	Baryon-like	10^{-10}–10^{-20} s

Table 5: Partial quantum-number and lifetime estimates for GTE-P1 through P11. Charge assignments use c-state matching (structural), Möbius product parity $\mu(a)\mu(b)\mu(c)$ (heuristic), or sector analogy (speculative). Lifetime classes: μs–stable = detector-stable or longer; intermediate values = displaced vertex or prompt. All assignments are provisional.

Möbius-product parity. For novel c-state predictions, the product $\mu(a)\cdot\mu(b)\cdot\mu(c)$ (where μ is the Möbius function) provides a structural parity proxy: +1 suggests lepton-like (even parity), -1 quark-like (odd parity), 0 composite or indeterminate. Results: GTE-P1 yields +1 (lepton-like); GTE-P8 yields -1 (quark-like). GTE-P2, P6, P9 yield 0 (composite factor present—sector indeterminate).

GTE-P7 quantum numbers (new, DS-B). The mirror-branch triple $(1, 73, 2137;\ g{=}1)$ has been assigned quantum numbers via the Braid Atlas [4]:

- $Q = 0$ (neutral): winding number $W_g = 0$ since the mirror duality is an internal GTE symmetry, not an SM gauge transformation.
- Color: singlet (single-strand sector, $a = 1$).
- Spin: $\frac{1}{2}$ (Dirac fermion; odd crossing parity).

Lean certificates: `mirror_triple_residue` (proves $2137 \bmod 73 = 20$ — same residue as canonical lepton triple), `mirror_prime_2137` (zero sorry, [9]). Supporting code: `papers/02_GTE_spectrum/mirror_branch_quantum_numbers.py`.

Claim grade: [B] bridge — from braid-atlas rules; now formally derived in `BraidAtlas.ChargeTheorem` (`gte_p7_electric_charge_zero`, zero `sorry`, 1 disclosed axiom `mirror_winding_number_zero`, [9]).

Lifetime derivation pathway. With quantum numbers now assigned for GTE-P7 (above), step (1) is complete. Full lifetime prediction further requires: (2) kinematic analysis of open decay channels, and (3) coupling-strength estimates from UGP selection rules. The lifetime classes in Table 5 are upper-bound estimates based on mass range and sector analogy only.

Future work. The trajectory-path multiplicity theorem (§6.3) provides a natural framework for quantum-number assignment: triples that differ only in path but share (a, b, c) should inherit the charge and sector quantum numbers of the canonical SM assignment, with potential modifications from the path-dependent parity sequence. Formalizing this within the Lean 4 machine-checked UGP framework [9] is a concrete next step.

C Reproducibility

Step 1: Discovery run (∼25 min, 14 workers).

```
python Verifier_discovery_engine_v4.py run \
  --mode discover_new \
  --preset comprehensive_gte_strict_search \
  --max-new-particles 25000000 \
  --output-dir discovery_runs/discovery_run_v4_<date>
```

Verify `candidates.csv` SHA-256 matches the value in Data Availability.

Step 2: Analytics (∼25 min, 14 workers).

```
python Verifier_discovery_advanced_particle_analysis.py \
  --csv <run>/candidates.csv \
  --mass_cutoff 173000.0 --max_curves 8 \
  --trials 5000 --bootstrap_n 200 --nperm 800 \
  --do_lifetime --do_surface --do_consensus \
  --fit_law_family --with_hinge2 --plot_heatmaps \
  --max_workers 14
```

Step 3: Paper figures.

```
python generate_paper_figures.py
```

D Artifact Manifest

All machine-readable artifacts accompanying this paper are listed in `PROVENANCE.md` in the paper directory. Key files:

- `candidates.csv` — 1,000,035 candidate particles (primary result)
- `law_family.json` — B, D parameters
- `oscillation_fdr_mass.json` — oscillation analysis with FDR
- `surfaces.json` — multivariate surface fits
- `consensus_scores.csv` — 23,450 cross-validated anchor candidates
- `calibration_diagnostics.json` — PCHIP calibrator details
- `settings.json` — discovery run parameters
- `manifest.json` — analytics run metadata and CSV hash